Role of micropores in TENG charge trapping

"What role do micropores play in charge trapping in TENGs" *Generated by LeapSpace*

Micropores mainly act as **additional physical trap sites** in TENGs: they increase internal surface area, create more dipole/interface regions, and help hold triboelectric charges longer, but if the microporosity becomes excessive or too conductive, the trapping benefit drops and output can fall. [89] [90] [91] [92] [93] [110]

What micropores do in charge trapping

Micropore role	Effect on charge trapping	TENG impact
More pore walls and edges	Provide more physical sites where electrons can be trapped and retained [91] [110]	Higher surface charge density and better output [89] [110]
Higher internal surface area	Increases the amount of interfacial area available for induced charge storage [92] [93] [94]	Stronger charge accumulation and longer retention [92] [94]
Microcapacitor-like domains	Local dielectric heterogeneity helps store and stabilize charge [92] [93]	Improved charge trapping and capacitance [92] [93]
Too many / too dense micropores	Can thin sidewalls, weaken mechanical recovery, or create leakage/conductive pathways [92] [110]	Output eventually decreases [92] [110]

Bottom line

For TENGs, **micropores are beneficial mainly as charge-trapping features**: they trap electrons at pore surfaces, edges, and internal interfaces, which slows charge decay and boosts surface potential. [89] [95] [110] The best performance is usually achieved with **controlled microporosity**, not the maximum possible micropore density, because too much porosity can introduce structural weakness or unwanted conductive pathways. [92] [110]

Confidence level: **High** — multiple independent TENG studies consistently show micropores improving charge trapping and charge density, with a clear caveat that over-porosity can reverse the benefit.

References

89. Wu, J., Li, Y., Song, A., (...), Xu, J. (2025) High-performance triboelectric nanogenerator based on mangosteen peel-derived carbon for energy harvesting and self-powered sensing. *Chemical Engineering Journal*. https://doi.org/10.1016/j.cej.2025.166516

90. Pan, Y., Li, M., Huang, T., (...), Qin, A. (2023) Dynamic co-irradiation techniques: A new method improving the electrical output of TENG by optimizing the charge capture capacity of polypropylene membrane. *Applied Materials Today*. https://doi.org/10.1016/j.apmt.2023.101979

91. Edi, K.K., Mazida, I., Nadhifah, F.R., (...), Jamaluddin, A. (2026) Recent advances in carbon-based materials for triboelectric nanogenerators (TENGs): An integrated bibliometric and systematic review. *Next Materials*. https://doi.org/10.1016/j.nxmate.2026.102648

92. Wang, Y., Gui, Y., He, S., Yang, J. (2023) MOF-derived NiO/N-MWCNTs@PVDF film and MXene/Carbon/Ecoflex electrode for enhanced hybrid nanogenerator towards environmental monitoring. *Composites Part A: Applied Science and Manufacturing*. https://doi.org/10.1016/j.compositesa.2023.107692

93. Rana, S.M.S., Rahman, M.T., Zahed, M.A., (...), Park, J.Y. (2022) Zirconium metal-organic framework and hybridized Co-NPC@MXene nanocomposite-coated fabric for stretchable, humidity-resistant triboelectric nanogenerators and self-powered tactile sensors. *Nano Energy*. https://doi.org/10.1016/j.nanoen.2022.107931

94. https://www.sciencedirect.com/science/article/pii/S2211285522000064

95. Salauddin, M., Rana, S.S., Sharifuzzaman, M., (...), Park, J.Y. (2022) Laser-carbonized MXene/ZiF-67 nanocomposite as an intermediate layer for boosting the output performance of fabric-based triboelectric nanogenerator. *Nano Energy*. https://doi.org/10.1016/j.nanoen.2022.107462

110. Cheng, E., Zhang, Z., Zhang, G., (...), Hu, N. (2023) A triboelectric nanogenerator coupled with internal and external friction for gesture recognition based on EHD printing technology. *Nano Energy*. https://doi.org/10.1016/j.nanoen.2023.108357